

SULTAN MAHMUD SAJAL

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RESEARCH SUMMARY

Sultan Mahmud Sajal is a computer systems researcher and software engineer whose work advances the methodology of systems research itself. Specifically, his work focuses on how cloud and AI infrastructure systems are evaluated, benchmarked, and scaled. His NSF-funded PhD research at Penn State (grants 1717571, 1909004, 2122155, 2239291) produced novel trace-driven experimentation techniques adopted in both academia (30+ citations) and production systems (Microsoft Azure). His publicly available tools (e.g., TraceUpscaler on GitHub) are used by researchers worldwide. His industry work at Meta and NetApp applies these principles to large-scale infrastructure challenges with measurable national impact, including \$100M/year in data center operational savings. His research is published in the field's most selective venues (OSDI, EuroSys, ACM TOCS) and he has presented his work at all three venues.

EDUCATION

Ph.D., Computer Science and Engineering

08/21/2018 - 08/10/2024

The Pennsylvania State University, PA, USA

Thesis: Improving the Fidelity of Trace-Driven Experiments in Cloud Computing Systems

Developed foundational methodologies for realistic, reproducible experimentation in cloud systems research

Produced 4 publications across OSDI, EuroSys (×2), and ACM TOCS

B.Sc., Computer Science and Engineering

02/16/2013 - 09/13/2017

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

Technical Scholarship recipient

PUBLICATIONS

- **Sultan Mahmud Sajal**, L Marshall, B Li, S Zhou, A Pan, K Mellou, D Narayanan, T Zhu, D Dion, T Moscibroda, I Menache. **“Kerveros: Efficient and Scalable Cloud Admission Control.”**. 17th USENIX Symposium on Operating Systems Design and Implementation, (OSDI 2023). — Deployed in Microsoft Azure production; achieved 99.9%+ availability and 95%+ resource utilization in global scale. **13 citations**
- **Sultan Mahmud Sajal**, T Zhu, B Urgaonkar, S Sen. **“TraceUpscaler: Upscaling Traces to Evaluate Systems at High Load.”**. Nineteenth European Conference on Computer Systems, (EuroSys 2024). — **Best Runner-Up Paper Award**. 7 citations.
- **Sultan Mahmud Sajal***, R Hasan*, T Zhu, B Urgaonkar, S Sen. **“TraceSplitter: A New Paradigm for Downscaling Traces.”**. Proceedings of the Sixteenth European Conference on Computer Systems, (EuroSys 2021). **Equal Contribution*. — 9 citations.
- **Sultan Mahmud Sajal**, Md Salman Estyak, Rubaba Hasan, T Zhu, B Urgaonkar, S Sen. **“TraceScaler: A Framework for Scaling Load in Real-World Traces for System Evaluation.”**. ACM Transactions on Computer Systems 43.4 (2025): 1-31. (*Invited Paper*) — Comprehensive framework generalizing trace scaling for cloud systems. 1 citation.
- **Sultan Mahmud Sajal**. **“Improving the Fidelity of Trace-Driven Experiments in Cloud Computing Systems.”**. *PhD Dissertation*, The Pennsylvania State University, (2024).

RESEARCH EXPERIENCE

The Pennsylvania State University (University Park, PA) - Graduate Assistant

08/21/2018 - 07/31/2024

- **Ph.D. Thesis:** Improving the Fidelity of Trace-Driven Experiments in Cloud Computing Systems.
- **Project:** TraceUpscaler — Upscaling Cloud Workload Traces
 - Developed novel algorithms to upscale real-world cloud traces to higher loads, enabling system evaluation under stress conditions not available in production.
 - Winner of Best Runner-Up Paper at EuroSys 2024.
 - Work addresses a fundamental limitation in cloud systems research, enabling more accurate benchmarking of future infrastructure.
- **Project:** TraceSplitter — Downscaling Cloud Workload Traces
 - Developed first principled framework for downscaling cloud traces while preserving statistical properties (arrival process, performance characteristics).
 - Enables realistic systems research using small-scale testbeds, lowering barriers to reproducible cloud research.
 - Published EuroSys 2021.
- **Project:** Reproducibility in Cloud Systems Experimentation
 - Investigated and quantified sources of non-reproducibility in trace-driven experiments.
 - Developed methodology and tooling for isolated, reproducible cloud experiments.
 - Directly addresses a systemic problem in how US research institutions validate systems work.

Microsoft Research, Microsoft Corporation (Redmond, WA) - Research Intern

05/31/2022 - 08/26/2022

- **Project:** Kerveros — Efficient and Scalable Cloud Admission Control (Azure)
 - Designed novel admission control system guaranteeing SLAs for both allocated and reserved cloud resources while maximizing global resource utilization.

- Achieved 99.9%+ availability and 95%+ resource utilization in Azure production deployment.
- Published OSDI 2023, one of the highest-impact systems research venues.

Gray Systems Lab, Microsoft Corporation (Redmond, WA) - Research Intern

05/24/2021 - 08/20/2021

- **Project:** Synthetic Benchmark Generation for Azure HDInsight
 - Reconstructed realistic Spark workloads from real-world query traces, generating synthetic datasets that preserve statistical properties while maintaining privacy.
 - Addressed the need for reproducible, privacy-preserving cloud benchmarks.

INDUSTRY EXPERIENCE

NetApp, Inc. (San Jose, CA) - Software Engineer 3

01/05/2026 - Present

- Optimizing performance of AIDE (AI Data Engine), a distributed platform for ingesting, cataloging, and serving enterprise storage data for AI workloads.
- Applying research expertise in distributed systems performance analysis and workload characterization to enable enterprise AI infrastructure

Meta Platforms, Inc. (Menlo Park, CA) - Performance and Capacity Engineer

05/06/2024 - 02/01/2026

- Optimized rack power budgets for multiple AI inference hardware programs, yielding \$100M/year in data center OpEx savings; demonstrating research-grounded methodology applied at national scale.
- Drove cross-functional collaboration to increase GPU power limits for recommendation services, achieving 7% reduction in GPU capacity usage (\$400K/year benefit) without additional power budget.
- Delivered power budgets for flash storage racks supporting 7.3EB (GenAI training) and 9.9EB (MySQL/services).
- Redesigned the Power Sufficiency Simulator pipeline, reducing Thrift service dependencies by 77%.
- Developed machine-level power estimation method improving power capping efficiency by 11%.
- Developed power-device-level estimation achieving 99%+ accuracy (eliminating 0.9%–6% error).
- Improved idle host power estimation accuracy by up to 31%.

REVE Systems (Dhaka, Bangladesh) - Junior Software Engineer

10/02/2017 - 07/10/2018

- Built a prototype machine translation system supporting Bangla and 25 languages
- Developed NLP platform prototype (machine translator, spell checker) for Bangla language

SERVICE AND PUBLIC CONTRIBUTIONS

- **Peer Reviewer**
 - ACM Symposium on Cloud Computing (SoCC).
- **Conference Presentations**
 - "Kerveros: Efficient and Scalable Cloud Admission Control" — USENIX OSDI 2023, Boston, MA (July 2023)
 - "TraceUpscaler: Upscaling Traces to Evaluate Systems at High Load" — EuroSys 2024, Athens, Greece (April 2024)
 - "TraceSplitter: A New Paradigm for Downscaling Traces" — EuroSys 2021, Online (April 2021)
- **Open-Source Contributions**
 - TraceUpscaler — publicly available at GitHub (<https://github.com/smsajal/TraceUpscaler>), with Zenodo archive DOI: 10.5281/zenodo.10042017
 - TraceSplitter - publicly available on GitHub (<https://github.com/smsajal/TraceSplitter>)
 - TraceScaler — publicly available on GitHub (<https://github.com/smsajal/TraceScaler>)
 - Kerveros - publicly available dataset on GitHub (<https://github.com/Azure/AzurePublicDataset>)

TECHNICAL SKILLS

Languages: Python, Java, C++, Scala, R

Frameworks/Tools: Pandas, Apache Spark, Docker, Kubernetes, Git

Cloud Platforms: AWS, Azure

Databases: MySQL, PostgreSQL, Presto

Specialized Areas: Distributed Systems, Systems Benchmarking, Power Budget Optimization, Performance Analysis, Workload Modeling

HONORS AND AWARDS

- **Best Runner-Up Paper Award**, European Conference on Computer Systems, (EuroSys) 2024
- **Student Grant**, European Conference on Computer Systems, (EuroSys) 2024
- **Student Grant**, USENIX Symposium on Operating Systems Design and Implementation, (OSDI) 2023

- **Registration Grant**, ACM SIGMETRICS 2021
- **Student Grant**, USENIX Symposium on Operating Systems Design and Implementation, (OSDI) 2020
- **Student Travel Grant**, ACM Symposium on Cloud Computing (SoCC) 2019
- **Student Travel Grant**, ACM Symposium on Cloud Computing (SoCC) 2018
- **Technical Scholarship**, Bangladesh University of Engineering and Technology 2018